# Exponentials & Polynomials

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01 • 20 FRQs + 25 MCQs • Difficulty Progression • Cross-Lesson Synthesis

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • [FRQ] • Target: Laws of Exponents & Fractional Powers

Simplify each expression using exponent and radical laws, assuming $x > 0$ and $y > 0$:

**a** $\frac{(2x^3y^{-2})^3}{4x^5y^{-4}}$

**b** $(27x^6)^{2/3} \cdot x^{-1}$

**2** [Foundational] • [FRQ] • Target: Conjugates & Simplifying Radicals

Rationalize the denominator and simplify $\frac{12}{3-\sqrt{5}}$ into the form $a + b\sqrt{5}$.

**3** [Foundational] • [FRQ] • Target: Polynomial Multiplication & Degree

Expand and simplify $(2x - 3)(x^2 + 4x - 5)$, and state the degree and leading coefficient of the resulting polynomial.

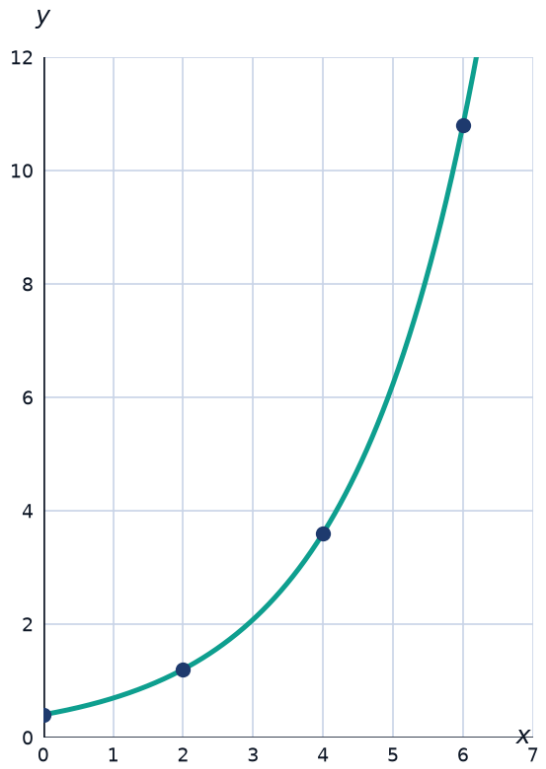
**4** [Foundational] • [FRQ] • Target: Polynomial Remainder Theorem

Use the Remainder Theorem to find the remainder when $P(x) = 2x^3 - 5x^2 + 3x - 7$ is divided by $x - 2$.

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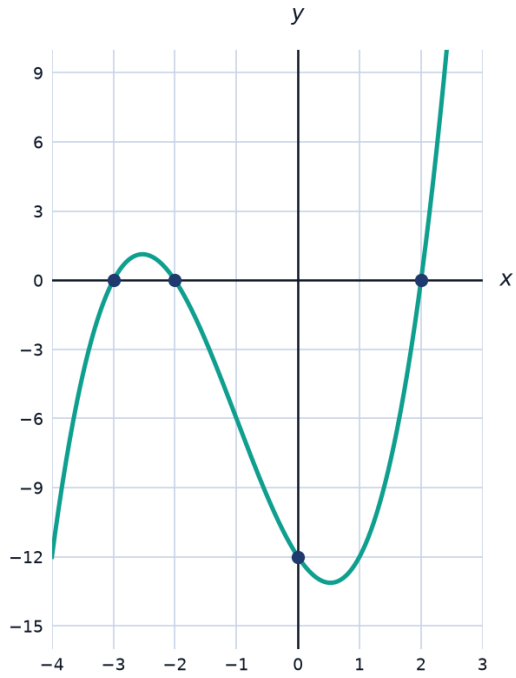
**5** [Foundational] • [FRQ] • Target: Exponential Growth Modeling

A bacterial culture starts with $400$ cells and triples every $2$ hours. Write an exponential model $N(t)$ for the population after $t$ hours, and calculate the population after $6$ hours.




**6** [Foundational] • [FRQ] • Target: Factoring Cubic Polynomials by Grouping

Factor the cubic polynomial $P(x) = x^3 + 3x^2 - 4x - 12$ completely by grouping and list all three real zeros.




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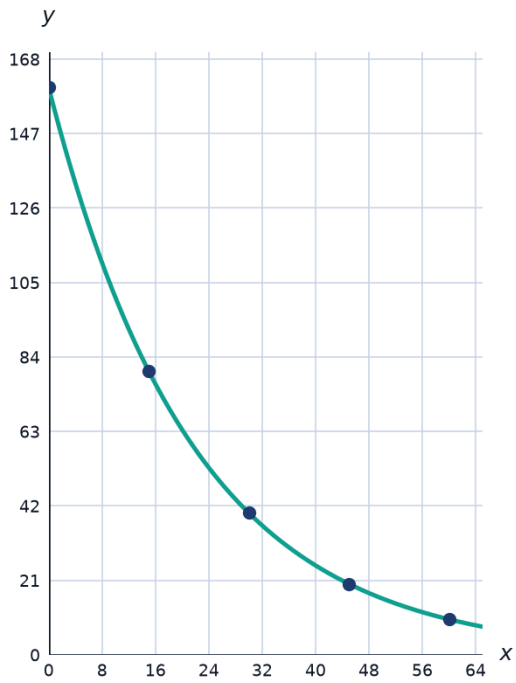
**7** [Foundational] • [FRQ] • Target: Solving Exponential Equations with Common Bases

Solve the exponential equation $27^{2x-1} = 9^{x+2}$ by rewriting both sides with a common base.

## Part I — Free-Response Questions (FRQs): Intermediate Modeling (8–14)

**8** [Intermediate] • [FRQ] • Target: Half-Life & Exponential Decay

A radioactive isotope has a half-life of $15$ days. If a sample initially has a mass of $160\,\text{mg}$, write the decay formula $M(t)$ after $t$ days and find how many days it takes for the sample to decay to $10\,\text{mg}$.




**9** [Intermediate] • [FRQ] • Target: Radical Equations & Extraneous Solutions

Solve the radical equation $\sqrt{3x+7} - x = 1$ and verify which candidate root(s) are extraneous.

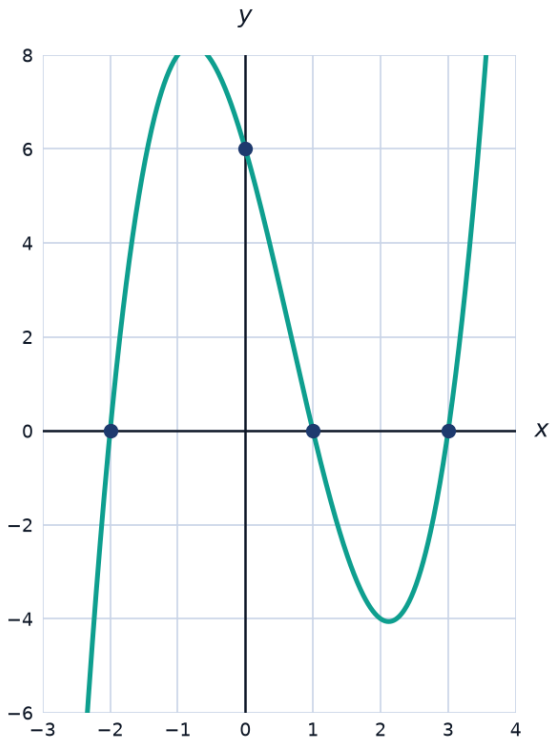
**10** [Intermediate] • [FRQ] • Target: Factor Theorem & Polynomial Division

Given that $x + 2$ is a factor of $P(x) = 2x^3 + kx^2 - 7x - 6$, find the constant $k$ and factor $P(x)$ completely.

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**11** **[Intermediate]** • **[FRQ]** • **Target: Polynomial Zeros, Multiplicity & Sign Analysis**

The cubic polynomial $f(x) = (x + 2)(x - 1)(x - 3)$ is graphed below. State its $x$-intercepts, its $y$-intercept, and the intervals on which $f(x) > 0$.




**12** **[Intermediate]** • **[FRQ]** • **Target: Compound Interest & Effective Rate Conversion**

An investment of $5{,}000$ dollars earns $8\%$ annual interest compounded quarterly. Write the expression for its value $A(t)$ after $t$ years, and find the effective annual percentage yield (APY) rounded to two decimal places.

**13** **[Intermediate]** • **[FRQ]** • **Target: Binomial & Cubic Algebraic Identities**

If $x - \frac{1}{x} = 4$, compute the exact value of $x^2 + \frac{1}{x^2}$ and $x^3 - \frac{1}{x^3}$.

**14** **[Intermediate]** • **[FRQ]** • **Target: Rewriting Exponential Time Scales**

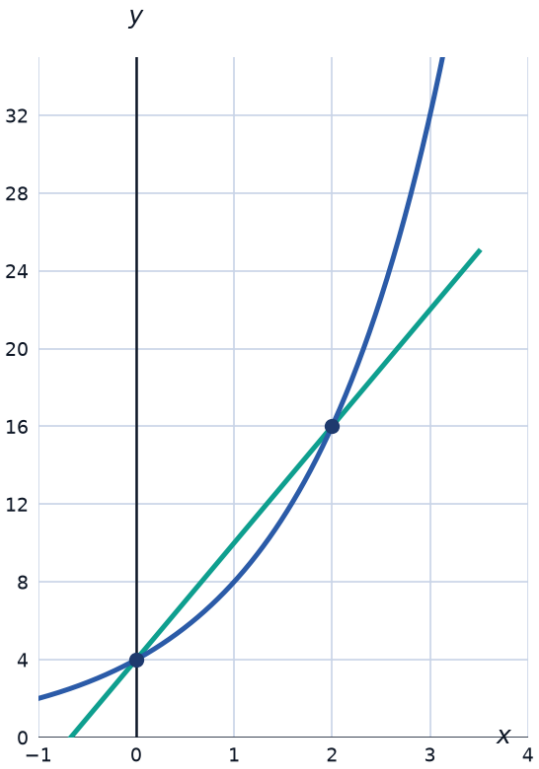
Rewrite the monthly growth model $V(t) = 1200(1.44)^{t/2}$ (where $t$ is in years) in the form $V(t) = 1200(1 + r)^t$ and state the annual percentage growth rate $r$.

## Part I — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • [FRQ] • Target: Linear vs. Exponential Growth Comparison • Cross-Lesson: Topic 1.01 (Linear Models & Rate Comparison)

Compare the linear function $L(x) = 6x + 4$ and the exponential function $E(x) = 2^{x+2}$ at $x = 0, 1, 2, 3, 4$. Find all integer solutions to $L(x) = E(x)$ and state for which $x$ we have $E(x) > L(x)$.



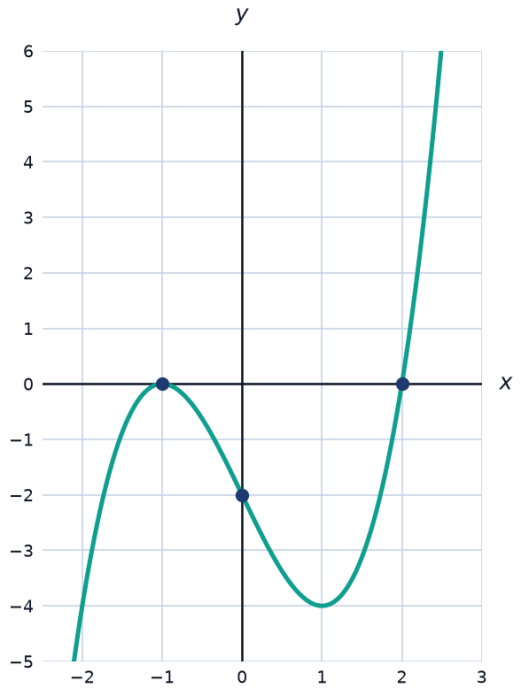

**16** [Challenging] • [FRQ] • Target: Polynomial Division with Quadratic Divisors • Cross-Lesson: Topic 2.01 (Systems of Linear Equations)

A polynomial $P(x)$ leaves a remainder of $5$ when divided by $x - 1$ and a remainder of $-3$ when divided by $x + 3$. Find the linear remainder $R(x) = ax + b$ when $P(x)$ is divided by $(x - 1)(x + 3)$.

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**17** [Challenging] • [FRQ] • Target: Multiplicity & Tangency of Polynomial Graphs • Cross-Lesson: Topic 3.01 (Quadratic Tangency)

Let $f(x) = (x+1)^2(x-2)$. Find the coordinates of all $x$- and $y$-intercepts, identify at which intercept the graph is tangent to the $x$-axis vs. crosses it, and find $f(3)$.




**18** [Challenging] • [FRQ] • Target: Systems of Exponential Equations • Cross-Lesson: Topic 2.01 (Linear Systems)

Solve the system of exponential equations $2^{x+y} = 32$ and $3^{2x-y} = 81$, and evaluate $x^2 + y^2$.

**19** [Challenging] • [FRQ] • Target: Scaling Laws with Rational Exponents • Cross-Lesson: Topic 8.01 (3D Solids) & Topic 5.01 (Percentages)

A cube has surface area $S = 6s^2$ and volume $V = s^3$. Express the volume $V$ as a function of surface area $S$ using rational exponents, and find the percentage increase in $V$ when $S$ is multiplied by $4$.

**20** [Challenging] • [FRQ] • Target: Dual-Radical Equations • Cross-Lesson: Topic 3.01 (Quadratics)

Find all real solutions to $\sqrt{x+5} + \sqrt{x-3} = 4$.

## Part II — Multiple-Choice Questions (MCQs): Foundational Concepts (21–28)

**21** [Foundational] • [MCQ] • Target: Converting Radicals to Rational Exponents

For $x > 0$, which expression is equivalent to $\sqrt[3]{x^5} \cdot \sqrt{x^3}$?

**a.** $x^{8/5}$

**b.** $x^{5/2}$

**c.** $x^{15/6}$

**d.** $x^{19/6}$

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**22** [Foundational] • [MCQ] • Target: Evaluating Exponential Expressions

If $3^{2x} = 81$, what is the value of $3^{x+1}$?

**a.** 9

**b.** 18

**c.** 27

**d.** 84

**23** [Foundational] • [MCQ] • Target: Polynomial Subtraction & Coefficient Sums

When $(3x^2 - 4x + 5) - (x^2 + 6x - 2)$ is written in the form $ax^2 + bx + c$, what is the value of $a + b + c$?

**a.** $-1$

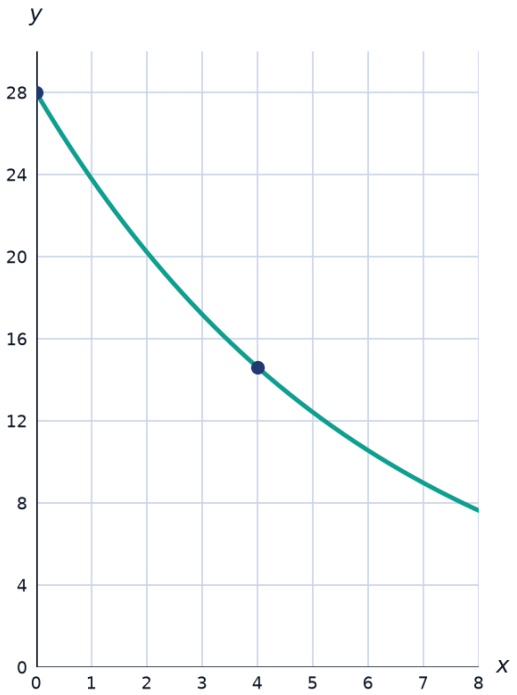
**b.** 3

**c.** 7

**d.** 11

**24** [Foundational] • [MCQ] • Target: Exponential Decay Factor

A car purchased for $28{,}000$ dollars depreciates in value by $15\%$ each year. Which function models the car's value $V(t)$, in dollars, $t$ years after purchase?




**a.** $V(t) = 28000(0.15)^t$

**b.** $V(t) = 28000(0.85)^t$

**c.** $V(t) = 28000(1.15)^t$

**d.** $V(t) = 28000 - 0.15t$

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**25** [Foundational] • [MCQ] • Target: Solving Basic Radical Equations

If $\sqrt{2x-3}=5$, what is the value of $x$?

**a.** 4

**b.** 11

**c.** 14

**d.** 28

**26** [Foundational] • [MCQ] • Target: Difference of Cubes Factorization

Which of the following is a factor of $x^3-27$?

**a.** $x+3$

**b.** $x^2-3x+9$

**c.** $x^2+3x+9$

**d.** $x^2-9$

**27** [Foundational] • [MCQ] • Target: Intercepts of Shifted Exponential Functions

What is the $y$-intercept of the exponential function $f(x)=5(2)^{x-1}-3$?

**a.** $(0,-0.5)$

**b.** $(0,2)$

**c.** $(0,5)$

**d.** $(0,7)$

**28** [Foundational] • [MCQ] • Target: Identifying Exponential Tables

Which table of values represents an exponential function rather than a linear or quadratic function?

**a.** $(1,3),(2,6),(3,9),(4,12)$

**b.** $(1,3),(2,6),(3,12),(4,24)$

**c.** $(1,1),(2,4),(3,9),(4,16)$

**d.** $(1,5),(2,7),(3,11),(4,17)$

## Part II — Multiple-Choice Questions (MCQs): Intermediate Exam Practice (29–37)

**29** [Intermediate] • [MCQ] • Target: Exponent Base Unification

If $\frac{8^x}{2^y}=16$, what is the value of $3x-y$?

**a.** 2

**b.** 3

**c.** 4

**d.** 8

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**30** [Intermediate] • [MCQ] • Target: Finding Coefficients with the Factor Theorem

When the polynomial $P(x) = x^3 - 4x^2 + kx + 6$ is divided by $x - 3$, the remainder is $0$. What is the value of $k$?

**a.** $-1$

**b.** $1$

**c.** $2$

**d.** $3$

**31** [Intermediate] • [MCQ] • Target: Interpreting Fractional-Period Exponential Models

The population of a town is modeled by $P(t) = 18000(1.04)^{t/3}$, where $t$ is the number of years after $2020$. Which statement best interprets this model?

**a.** The population increases by $4\%$ every year.

**b.** The population increases by $4\%$ every $3$ years.

**c.** The population increases by $3\%$ every $4$ years.

**d.** The population increases by $1.04\%$ every $3$ years.

**32** [Intermediate] • [MCQ] • Target: Matching Polynomial Coefficients

If $(ax + 2)(bx - 3) = 12x^2 + cx - 6$ for all $x$, and $a + b = 7$, what are the two possible values of $c$?

**a.** $-6$ and $-1$

**b.** $-4$ and $1$

**c.** $-5$ and $5$

**d.** $-2$ and $2$

**33** [Intermediate] • [MCQ] • Target: Direct Fractional Power Scaling

For $x > 0$, if $x^{3/4} = 27$, what is the value of $x^{1/2}$?

**a.** $3$

**b.** $9$

**c.** $27$

**d.** $81$

**34** [Intermediate] • [MCQ] • Target: Constructing Polynomials from Roots & Multiplicity

The polynomial $p(x)$ of degree $4$ has a double root at $x = -2$, a simple root at $x = 1$, a simple root at $x = 3$, and $p(0) = 12$. What is $p(2)$?

**a.** $-16$

**b.** $-8$

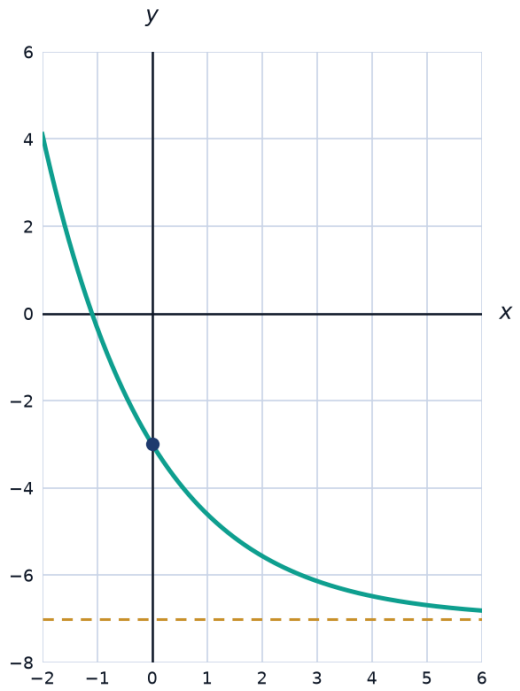
**c.** $8$

**d.** $16$

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**35** **[Intermediate]** • **[MCQ]** • **Target: Asymptotes of Exponential Functions**

What is the horizontal asymptote of the graph of $f(x) = 4(0.6)^x - 7$ in the $xy$-plane?




**a.** $y = -7$

**b.** $y = -3$

**c.** $y = 0$

**d.** $y = 4$

**36** **[Intermediate]** • **[MCQ]** • **Target: Combining & Re-Expressing Radicals**

If $\sqrt{50} + \sqrt{18} = \sqrt{k}$, what is the value of $k$?

**a.** 68

**b.** 96

**c.** 128

**d.** 144

**37** **[Intermediate]** • **[MCQ]** • **Target: Sum of Cubes Identity**

If $x^3 + y^3 = 72$ and $x + y = 6$, what is the value of $xy$?

**a.** 6

**b.** 8

**c.** 10

**d.** 12

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## Part II — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • [MCQ] • Target: Exponential Equations Reducible to Quadratics • Cross-Lesson: Topic 3.01 (Quadratics)

What is the sum of all real solutions to $9^x - 12 \cdot 3^x + 27 = 0$?

**a.** 3

**b.** 4

**c.** 12

**d.** 27

**39** [Challenging] • [MCQ] • Target: Log-Linearized Exponential Models • Cross-Lesson: Topic 1.01 (Linear Equations) & Topic 6.01 (Data Modeling)

In a semi-log plot where $\log_{10}(y)$ is plotted on the vertical axis against $x$ on the horizontal axis, the data points lie on the straight line $\log_{10}(y) = 2x + 3$. Which function gives $y$ in terms of $x$?

**a.** $y = 1000(100)^x$

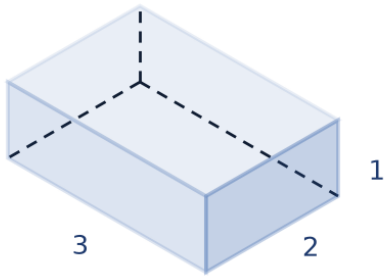
**b.** $y = 3(10)^{2x}$

**c.** $y = 100(1000)^x$

**d.** $y = 2x + 1000$

**40** [Challenging] • [MCQ] • Target: Cubic Vieta's Formulas & 3D Surface Area • Cross-Lesson: Topic 8.01 (3D Solids & Surface Area)

The polynomial $P(x) = x^3 - 6x^2 + 11x - 6$ has three positive integer zeros that represent the lengths of the three edges of a rectangular prism. What is the surface area of the prism?




**a.** 6

**b.** 11

**c.** 22

**d.** 36

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**41** [Challenging] • [MCQ] • Target: Extraneous Root Filtering • Cross-Lesson: Topic 3.01 (Quadratics)

For how many real values of $x$ does $\sqrt{x+7} = x - 5$ hold?

**a.** Zero

**b.** One

**c.** Two

**d.** Three

**42** [Challenging] • [MCQ] • Target: Compound Percentage Scaling Across Multiples of Time • Cross-Lesson: Topic 5.01 (Percentages)

A savings account with continuous-style annual growth grows by $60\%$ over every $4$-year period. By what percentage does the account grow over an $8$-year period?

**a.** $120\%$

**b.** $140\%$

**c.** $156\%$

**d.** $160\%$

**43** [Challenging] • [MCQ] • Target: Polynomial Composition & Substitution • Cross-Lesson: Topic 10.01 (Function Composition)

If $P(x)$ is a polynomial such that $P(x^2 + 1) = x^4 + 6x^2 + 8$ for all real $x$, what is the value of $P(5)$?

**a.** 24

**b.** 35

**c.** 48

**d.** 63

**44** [Challenging] • [MCQ] • Target: Coordinate Geometry of Polynomial Secants/Tangents • Cross-Lesson: Topic 1.01 (Linear Equations) & Topic 7.01 (Triangle Area)

What is the area of the triangle formed by the $x$-axis, the $y$-axis, and the tangent line to $y = x^2$ at $(0, 0)$ shifted to pass through $(2, 4)$ with slope $4$?

**a.** 1

**b.** 2

**c.** 4

**d.** 8

**45** [Challenging] • [MCQ] • Target: Comparing Large Exponentials via GCD Exponents • Cross-Lesson: Topic 2.01 (Inequalities & Ordering)

If $a = 2^{60}$, $b = 3^{45}$, and $c = 5^{30}$, which inequality correctly orders $a$, $b$, and $c$?

**a.** $a < c < b$

**b.** $a < b < c$

**c.** $c < a < b$

**d.** $b < c < a$

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